Multipatch Perception and Knowledge-Guided Network for Semantic Segmentation of Large-Scale Remote Sensing Image

Yijie Zhang, Jian Cheng, *Member, IEEE*, Ziying Xia, Wenqi Yin, Siyu Liu, and Nyima Tashi

***Abstract*—Semantic segmentation of large-scale remote sensing (LSRS) images has always been a hot research topic. Recently, methods have been proposed that segment LSRS images by dividing them into small patches, which results in the neglect of interpatch connections during the segmentation process, thereby causing incomplete global semantic information. To alleviate this problem, we propose a multipatch perception and knowledge-guided network (MPKG-Net) for the semantic segmentation of LSRS images. The MPKG-Net has two significant characteristics: *1) multipatch perception:* Initially, MPKG-Net constructs a multipatch feature representation for patches relevant to the current segmented patch. Subsequently, the multipatch feature is further processed through the multipatch perception branch built upon the global feature selection block (GFSB). The extracted multipatch features are incorporated into all encoder stages of MPKG-Net, enabling the model to perceive abundant global semantic information; and *2) knowledge-guided:* to effectively utilize multipatch features and enhance local features, we employ a pretrained base model on LSRS image datasets to construct the knowledge-guided branch. This branch processes features from both the multipatch perception branch and the local feature capture branch through the knowledge-guided multipatch feature filtering module (KG-MFFM) and the knowledge-guided local feature enhancement module (KG-LFEM), respectively. Finally, all features are fused stage-by-stage within the multifeature fusion branch. Our proposed model has achieved excellent performance on multiple publicly available LSRS images semantic segmentation datasets. The code will be available at https://github.com/zhangyijie199703/MPKG-Net**

***Index Terms*—Feature enhancement, global feature, knowledge-guided, multipatch perception, semantic segmentation.**

I. Introduction

With the rapid advancement of remote sensing technology, large-scale remote sensing (LSRS) images have emerged as a pivotal tool for acquiring and analyzing geospatial information [1], [2], [3], [4]. LSRS images are not only extensively utilized in ground object recognition and classification but also undertake diverse processing tasks such as object detection, change detection, and three-dimensional reconstruction [5], [6], [7], [8], [9], [10]. Semantic segmentation, a fundamental computer vision task, aims to assign each pixel in an image to one of the predefined semantic categories, enabling fine-grained classification and annotation at the pixel level [11], [12], [13], [14], [15], [16], [17], [18]. This facilitates a deeper understanding of image content by computers. Semantic segmentation of LSRS images can achieve precise recognition and classification of ground objects, providing robust decision support for fields such as land resource utilization, environmental monitoring, and disaster assessment.

Earlier semantic segmentation methods relied heavily on geometric constraints, statistical rules, and hand-crafted object features, necessitating extensive prior knowledge to guide the segmentation process. Due to the lack of effective automatic feature learning capabilities, these methods exhibited lower segmentation accuracy in complex scenes. Leveraging the ability to automatically learn complex feature representations from vast amounts of data, deep learning-based methods [19] have been extensively adopted in the field of image semantic segmentation. FCNs [11] were pioneers in achieving end-to-end pixel-wise predictions. Subsequently, UNet [13] improved upon the FCN by introducing skip connections that linked corresponding encoding and decoding layers, thereby complementing potentially lost detailed features. Many subsequent works further expanded the receptive field based on the encoder–decoder architecture. For instance, PSPNet [20] employed parallel pooling kernels of different sizes to enlarge the receptive field, while DeepLabV3 [15] utilized parallel dilated convolutions for the same purpose. Furthermore, networks such as DANet [21] and CCNet [17] incorporated attention mechanisms to establish global dependencies.The above CNN-based methods are mostly limited to local receptive fields and cannot achieve the construction of global information of ground objects in LSRS images. In recent years, with the rising popularity of Transformers in the vision domain, numerous efforts have been made to apply them to semantic segmentation tasks. SETR [22] is an innovative semantic segmentation model that employs a Transformer architecture as the encoder, combined with a CNN decoder, effectively enhancing the performance of image segmenta-

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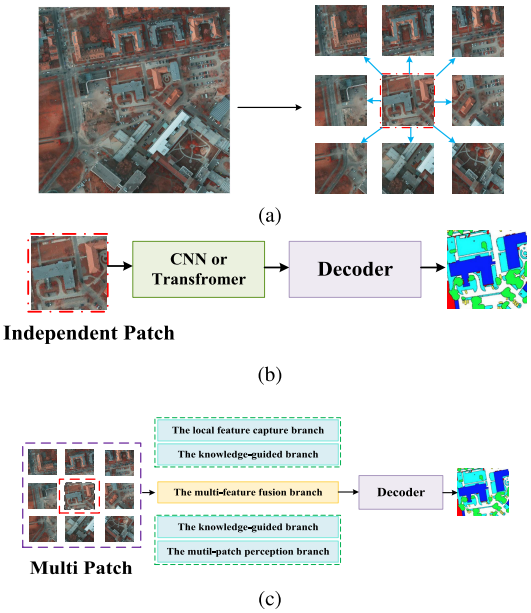


Fig. 1. (a) Each patch in the LSRS image does not exist independently; rather, it has semantic relationships with other patches. (b) Most other LSRS image semantic segmentation methods segment each patch independently, without considering the semantic relationships between them. (c) Our method improves LSRS image semantic segmentation performance through multipatch perception and knowledge guidance.

tion tasks. Mask2Former [23] is another image segmentation model based on a Transformer, which integrates masking techniques, self-attention mechanisms, and multiscale feature extraction to accurately segment target objects from input images. Transformer-based networks with self-attention mechanisms can better construct rich global semantic relationships within images, thereby further advancing the development of semantic segmentation.

Although Transformer-based methods can establish global dependencies through self-attention mechanisms, considering the ultrahigh resolution of LSRS images, most methods still only construct global relationships within independent patches, failing to truly establish global semantic dependencies between patches. Current methods crop them into patches and segment each independently, similar to natural image segmentation, as shown in Fig. 1(b). However, unlike natural images, LSRS images have more complex relationships among ground objects, involving long-distance dependencies with broader contextual information. Each patch may lack comprehensive semantic information. Some methods attempt to compress LSRS images into low-resolution images (e.g., 512×512), but this approach still loses a lot of spatial detail information, resulting in inaccurate localization and segmentation of many small objects. Some methods attempt to use multipatch perception. For example, Liu et al. [24] proposed a patch-wise semantic segmentation method. They segmented LSRS images into local patches to construct a patch-wise network and adopted a multiscale training strategy to achieve efficient and accurate semantic segmentation of LSRS images. In addition, Pang et al. [25] designed an LSRS image semantic segmentation framework from image patches to regions, which can effectively combine local and global information. Although the above-mentioned methods based on multipatch perception have made some progress, there are still challenges in efficiently utilizing information from multiple patches to further improve segmentation performance.

Currently, the field of remote sensing has witnessed the emergence of numerous diverse foundation models [26], [27]. These foundation models often possess complex structures and have undergone pretraining on LSRS datasets, learning domain-specific knowledge such as spectral features and spatial distribution patterns of ground objects. In previous studies, it has been demonstrated that utilizing domain-specific knowledge of the foundation model can improve the performance of various tasks in the domain. Zhang et al. [28] utilized visual foundation models such as grounding DINO and CLIP to automatically generate visual prompts like points or bounding boxes, guiding SAM for remote sensing semantic segmentation. This approach of integrating multiple pretrained foundation models endows the network with powerful knowledge priors. In addition, RSAM-Seg [29] utilizes the Adapter-Scale and Adapter-Feature modules to incorporate prior knowledge, such as high-frequency information and embedding features of remote sensing images as a guide into the SAM model, achieving semantic segmentation of remote sensing images without the need for manual prompts. However, how to leverage the domain-specific knowledge embedded in these foundation models to mine and construct global semantic information among multiple patches, thereby enhancing the performance of LSRS images' semantic segmentation, has become a key issue.

Based on the above analyses, we propose a multipatch perception and knowledge-guided network (MPKG-Net) for the semantic segmentation of LSRS images. The simple schematic of the MPKG-Net structure is shown in Fig. 1(c). The encoder of MPKG-Net comprises four branches: the knowledge-guided branch, the local feature capture branch, the multipatch perception branch, and the multifeature fusion branch. Specifically, the multipatch perception branch is responsible for extracting broader global semantic information from the multipatch features within other patches relevant to the current segmented patch. The local feature capture branch utilizes the ConvNeXtV2 [30], [31] to extract local features of ground objects. The knowledge-guided branch employs the remote sensing foundation model Multitask pretraining approach (MTP) [26] to capture and establish rich domain-specific knowledge features from the input images and utilizes the knowledge-guided multipatch feature filtering module (KG-MFFM) and the knowledge-guided local feature enhancement module (KG-LFEM) to achieve multipatch feature filtering and local feature enhancement, respectively, while the multifeature fusion branch integrates various types of features progressively and concatenates them during the decoding stage to produce the final prediction result. Overall, the core innovation of MPKG-Net lies in achieving high-precision segmentation of LSRS images through multipatch perception and knowledge guidance. The main contribution of this article is summarized as follows.

1) We propose a novel MPKG-Net for LSRS image semantic segmentation. This model surpasses the size of most

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current LSRS image semantic segmentation models, featuring approximately 500 million parameters. Leveraging its larger number of model parameters, along with a multipatch perception and knowledge-guided architecture, MPKG-Net achieves high-precision LSRS semantic segmentation.

2) We design two modules that can implement knowledge guidance. Specifically, the KG-MFFM is used for filtering and selecting global features in LSRS images, and the KG-LFEM is used for enhancing the representation of local features.
3) We design the global feature selection block (GFSB) and build the multipatch perception branch based on it to capture global semantic information between multiple patches.
4) We conduct extensive comparative experiments on ISPRS Vaihingen and Potsdam datasets, and the experimental results demonstrate that our proposed method outperforms other current state-of-the-art methods.

II. Related Works

A. Semantic Segmentation

In the realm of semantic segmentation, methods based on the CNN and Transformer currently constitute the two mainstream technological approaches. CNN exhibits excellent performance in extracting image features thanks to its local receptive field and weight-sharing properties. It can effectively capture the local spatial information within images and progressively extract features at various levels through multilayer convolution kernels. Transformers, in contrast, have garnered widespread attention due to the advantage of their self-attention mechanism in capturing long-range dependencies. This mechanism empowers the model to model the relationships among different regions in the image without being constrained by distance, thereby presenting a new perspective for semantic segmentation.

Currently, semantic segmentation algorithms based on CNN can be approximately categorized into three types: simple encoding and decoding networks (FCN [11], UNet [13], SegNet [12],...,), receptive field-increasing networks (PSPNet [20], DeepLabV3+ [15],...,), and attention mechanism networks (DANet [21], CCNet [17], OCRNet [32],...,). FCN is the first network to achieve end-to-end pixel-level prediction. It accepts input images of arbitrary size by replacing fully connected layers with convolutional layers. UNet is widely utilized in biomedical imaging applications since it connects the encoding and decoding layers of corresponding stages through a skip-connection structure, thereby enriching the representation of detailed information. As a simple and efficient architecture, many semantic segmentation methods are based on improvements to UNet, encompassing image segmentation across various fields. SegNet employs indexed maxpooling upsampling by recording pooling positions to improve both segmentation accuracy and computational efficiency. PSPNet utilizes pyramid pooling modules (PPM) to capture contextual information at different scales, while DeepLabV3+ uses atrous spatial pyramid pooling (ASPP) to expand the receptive field and capture multiscale contextual information. DANet combines the spatial attention mechanism and the channel attention mechanism to effectively capture global feature dependencies in images. CCNet significantly optimizes the computational efficiency of attention and achieves precise segmentation by calculating the spatial dependencies of horizontal and vertical pixels. OCRNet introduces an object-context representation mechanism, which enhances the model's comprehension of intricate scenes and boosts segmentation precision.

CNN-based methods are good at local feature extraction with mature architectures, high efficiency, and easy training. Transformer-based methods are superior in capturing long-distance dependencies for complex scenes. The Vision Transformer (ViT) [33], introduced by Dosovitskiy et al. in 2020, revolutionized image classification models by replacing traditional convolutional layers with a Transformer encoder. This innovation enabled the ViT to efficiently capture global representations. Building on ViT's success, the Swin Transformer [34] introduced hierarchical architectures and window-based self-attention mechanisms. These features overcame the computational challenges faced by the original Transformer in processing high-resolution images, allowing the Swin Transformer to achieve state-of-the-art results across various image classification datasets. In the realm of semantic segmentation, SETR [22] leveraged the ViT architecture to create a new type of segmentation model. By using a complete ViT structure as its backbone, SETR enhanced contextual information capture and feature expression, offering fresh perspectives for semantic segmentation tasks. Meanwhile, Segformer [35] emerged as a simple yet highly effective semantic segmentation model that harnessed Transformers for both efficiency and accuracy. It delivered real-time performance without compromising on accuracy. MaskFormer [23] represented a significant advancement in both semantic and instance segmentation tasks by introducing mask classification. This innovative framework simplified complex segmentation challenges and delivered remarkable performance. Topformer [36], tailored for mobile devices, introduced a lightweight approach to semantic segmentation. It featured a token pyramid module to capture local features and a semantics extractor to model global and local relationships, making it suitable for on-device deployment. Recently, SegNeXt [37] emerged as a state-of-the-art model that combined the strengths of both the CNN and Transformer. This hybrid approach leveraged the advancements in both paradigms to achieve superior segmentation performance. Moreover, the segment anything model (SAM) [38], with its powerful zero-shot generalization capability, supports various input prompts and is currently widely applied in the field of image segmentation.

B. LSRS Image Semantic Segmentation

LSRS images exhibit diverse ground object categories, large-scale variations, small interclass variances, and large intraclass variances. Consequently, many researchers have enhanced general image semantic segmentation methods to obtain approaches suitable for LSRS image semantic segmentation.

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Zhao et al. [39] proposed an LSRS semantic segmentation network based on an attention mechanism called SSAtNet. This network incorporates a pyramid attention pooling module to adaptively refine multiscale features and a pooling index correction module to recover fine-grained details. In the realm of semantic segmentation for remote sensing images, effectively segmenting the boundaries of ground objects is a pivotal issue. Li et al. [40] introduced a boundary-aware enhancement method for LSRS image segmentation. Specifically, this method emphasizes the boundary information of ground objects during feature extraction and designs an edge-supervised loss function to oversee the learning of edge features. Liu et al. [41] observed that most existing semantic segmentation methods solely utilize the Transformer in the encoding stage while continuing to rely on CNN in the decoding stage, potentially leading to inconsistent feature representations. Consequently, they proposed a Transformer-based encoding-decoding method named GLOTS. Zhang et al. [42] proposed a truncated compensation network (TCNet) for LSRS image semantic segmentation. This method compensates for truncated features and minimizes noise in local images through a guided feature extraction module, enhances the recognition of global semantic information through a related category semantic enhancement module, and facilitates local image semantic segmentation with extensive context information through a global-local context cross-fusion module. Ma et al. [43] proposed the FTransUNet semantic segmentation framework, which is capable of integrating multimodal remote sensing information. This method harnesses the strengths of both the CNN and ViT to thoroughly extract crucial semantic information from multimodal features, thereby establishing a solid foundation for precise segmentation. In addition, some methods [24], [25] use a strategy of multipatch perception to further enhance the segmentation challenges caused by the large size of remote sensing images. Pang et al. [25] designed a semantic segmentation framework suitable for high-resolution remote sensing images, which is based on a patch-to-region architecture and can effectively perceive weak texture regions in LSRS images.

Considerable advancements have been made in current LSRS image semantic segmentation methods. However, most of the current semantic segmentation methods for LSRS are not significantly different from those for natural images. These methods ignore the relationships between patches in LSRS images, resulting in incomplete global semantic information during the segmentation process. This leads to unclear segmentation of ground object boundaries, category confusion, and other issues.

C. Remote Sensing Foundation Model

With the advent of the big data era, the significance of foundation models has become increasingly prominent. In deep learning, foundation models represent a pivotal architecture that serves as the fundamental building block for various downstream tasks. The core design objective of these models is to learn universal features and representations from large-scale datasets. The "domain knowledge" of foundation models is the information learned from a vast amount of data within a specific domain, reflecting the data's internal laws, like data distribution patterns and element associations. This knowledge helps the model deal with tasks in that domain. Foundation models can capture data patterns and structures. They can be adjusted and extended for specific scenarios, offering strong support for complex tasks [27], [44].

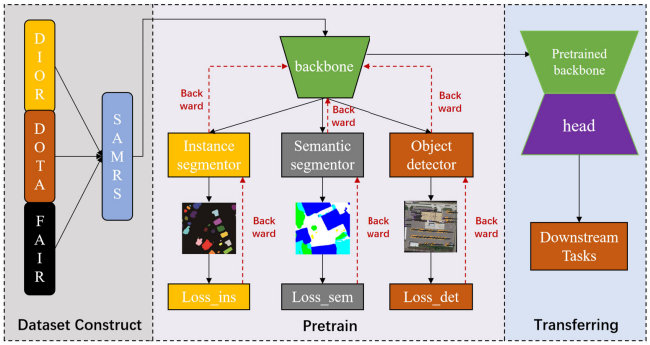


Fig. 2. MTP for remote sensing foundation models.

In the field of remote sensing, Hong et al. [45] addressed the research gap in spectral data for visual representation learning by creating a universal remote sensing foundation model named SpectralGPT. This model employs a novel three-dimensional generative pretrained Transformer architecture specifically designed to process spectral remote sensing images, aiming to enhance scene understanding in various applications. He et al. [46] proposed a foundation model adaptation (FMA) framework for multimodal remote sensing data classification without the need for training from scratch. By introducing cross-spatial interaction modules and cross-channel interaction modules to extract multimodal-specific representations, and further proposing the FMA2 method based on FMA, they effectively addressed the differences among multiple modalities using a coupling score function, thereby improving classification performance. Wang et al. first utilized the LSRS dataset MillionAID to pretrain the backbone and achieved excellent performance in multiple downstream remote sensing tasks [47]. Subsequently, they improved the self-attention mechanism in the ViT, proposing the novel attention mechanism tailored for ground object targets [48]. Based on this, they presented the first vision large model specifically for remote sensing image tasks, with a model size of 100 million parameters. In [49], by combining SAM with existing datasets for remote sensing object detection, they generated a large-scale dataset named SAMRS for remote sensing image semantic segmentation, which further promoted the development of networks for this task. In [26], an MTP for remote sensing foundation models was proposed, supporting both CNN and ViT foundation models with over 300 million parameters. The specific execution process of MTP is shown in Fig. 2. First, MTP constructs a large-scale dataset (SAMRS) by utilizing existing remote sensing datasets, namely DOTA, DOIR, and FAIR. Subsequently, pretraining is conducted on SAMRS for prevalent vision tasks, including object detection, semantic segmentation, and instance segmentation. Ultimately, the pretrained models can be applied to downstream tasks in remote sensing.

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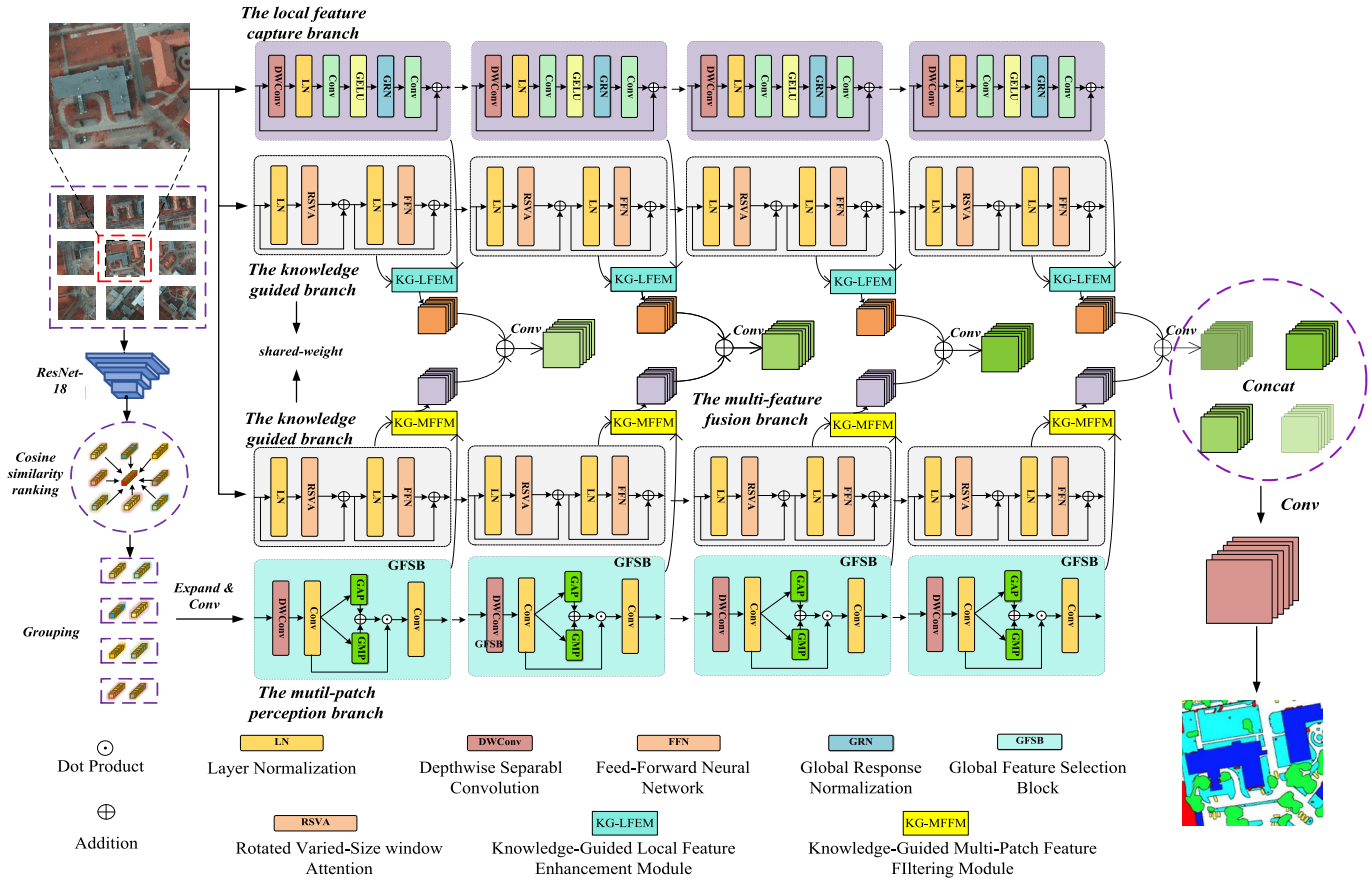


Fig. 3. Structure of the proposed MPKG-Net. The input of MPKG-Net not only includes the current patch image to be segmented, but also highly correlated patch images. The encoder of MPKG-Net consists of four branches: the local feature capture branch, the multipatch perception branch, the knowledge-guided branch, and the multifeature fusion branch.

In our proposed method, MTP is utilized as the knowledge-guided branch. MTP has been pretrained on LSRS datasets; the features encoded by it contain a substantial amount of information pertaining to the remote sensing domain. In this article, we term the features derived from MTP as knowledge features.

III. Methodology

A. MPKG-Net

A MPKG-Net for semantic segmentation of LSRS is proposed in this article. This method can achieve multipatch perception and utilize a knowledge-guided structure to filter multipatch features and enhance local features, thereby achieving efficient semantic segmentation of LSRS images. MPKG-Net mainly includes two key processes: multipatch perception and knowledge guidance. The overall structure of MPKG-Net is shown in Fig. 3.

The encoder of MPKG-Net comprises a shared-weight knowledge-guided branch, a local feature capture branch, a multipatch feature perception branch, and a multifeature fusion branch. Specifically, the knowledge-guided branch is constructed based on the remote sensing foundation model MTP, with its captured features being rich in domain knowledge. The local feature capture branch is built upon ConvNeXtV2, which can effectively extract the local features of ground objects, such as texture details. We establish the multipatch perception branch based on the GFSB to efficiently acquire multipatch features, realizing the in-depth perception of global semantic information. The multifeature fusion branch is used to fuse the features of each stage in the encoding process.

The patch that needs to be segmented is fed into both the knowledge-guided branch and the local feature capture branch, while the multipatch features G are directed to the multipatch perception branch. The construction of the multipatch features G will be described in detail in the next subsection. It is worth noting that the local feature capture branch and the multipatch perception branch are guided by a knowledge guidance branch that shares weight parameters for feature guidance. In addition, all branches have four consecutive encoding stages. At each stage of feature extraction, KG-LFEM enhances local features by leveraging knowledge features in both spatial and channel feature dimensions. Additionally, KG-MFFM guides multipatch features through knowledge features to achieve feature filtering and screening, thereby enabling efficient utilization of multipatch features. Subsequently, in the multifeature fusion branch, the enhanced local features and filtered multipatch features are summed to obtain the output for the current stage. The following shows the process of the first stage encoding:

$$F_{\mathrm{kg}}, F_{\mathrm{local}}^{\mathrm{init}}, F_{\mathrm{multipatch}} = \mathcal{B}_{\mathrm{kg}}(I_x), \mathcal{B}_{\mathrm{local}}(I_x)\, \mathcal{B}_{\mathrm{multipatch}}(G) \quad (1)$$

$$F_{\mathrm{local}}^{\mathrm{enhanced}} = \text{KG-LFEM}\left(F_{\mathrm{local}}^{\mathrm{init}}, F_{\mathrm{kg}}\right) \quad (2)$$

$$F_{\mathrm{multipatch}}^{\mathrm{filtered}} = \text{KG-MFFM}\left(F_{\mathrm{multipatch}}, F_{\mathrm{kg}}\right) \quad (3)$$

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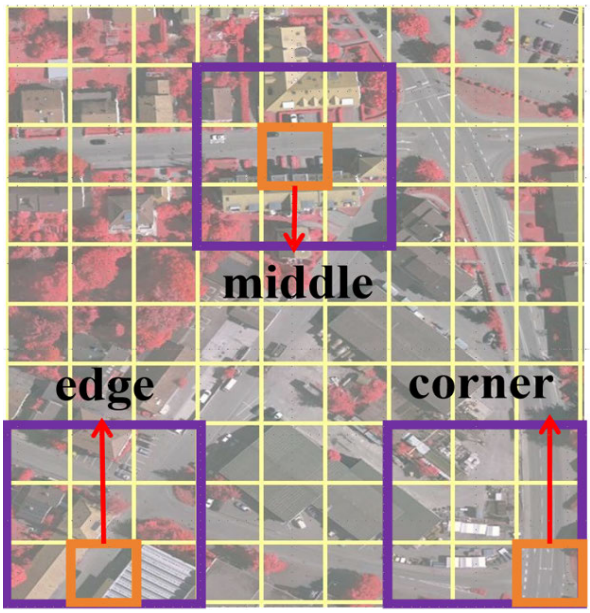


Fig. 4. Patches in LSRS images are not completely unrelated to each other. The semantic understanding of a single patch needs to rely on the contextual information of surrounding patches. We classify patches into three types based on their location in the LSRS image.

$$F_{\text{output}} = F_{\text{local}}^{\text{enhanced}} + F_{\text{multipatch}}^{\text{filtered}} \tag{4}$$

where $\mathcal{B}_{\text{kg}}$, $\mathcal{B}_{\text{local}}$, and $\mathcal{B}_{\text{global}}$ represent the knowledge-guided branch, the local feature capture branch, and the multipatch feature branch, respectively. F_{kg}, $F_{\text{local}}^{\text{init}}$, and $F_{\text{multipatch}}$ represent the knowledge feature, the local feature, and the multipatch feature, respectively. $F_{\text{local}}^{\text{enhanced}}$ and $F_{\text{global}}^{\text{filtered}}$ represent the enhanced local feature and filtered multipatch feature, respectively. F_{output} represents the output of the current stage of the multifeature fusion branch.

Finally, the outputs of the multifeature fusion branch from all stages are concatenated to obtain the final feature encoding result. The encoded features then undergo upsampling to produce the final output.

B. Multipatch Selection and the Processing Mechanism

Different from most current LSRS segmentation methods, the input of MPKG-Net not only includes the patch to be segmented but also other patches from the same large-scale image to achieve multipatch perception. In detail, considering the utilization of computational resources and the correlation between patches, we meticulously select eight patches that exhibit the highest correlation with the current patch to be segmented. Specifically, as shown in Fig. 4, we categorize the patch images to be segmented into three types: those located at the middle of the large-scale image, those situated along the edge, and those positioned at the corner. For each type of patch image, the segmentation process takes into account the additional patches within the purple bounding box to achieve multipatch perception and thus mine richer global semantic information.

The input size of all patches is $3 \times H \times W$. Therefore, the input of our model can be represented as I_x and $I_n = \{I_1, I_2, \ldots, I_8\}$. Among them, I_x is the patch image to be segmented, and I_n is the set of other related patches. First, we use a pretrained ResNet-18 on the ImagNet dataset to construct global feature vectors for all patches. Specifically, the input patches will pass through the pretrained ResNet-18 to obtain the encoded features F_x of I_x and the encoded features $F_n = \{F_1, F_2, \ldots, F_8\}$ of I_n. Subsequently, F_x and F_n will pass through global average pooling (GAP) to obtain global feature vectors G_x and $G_n = \{G_1, G_2, \ldots, G_8\}$. The dimensions of the global feature vectors in G_x and G_n are both $512 \times 1 \times 1$. The above process can be expressed as follows:

$$F_x = \text{ResNet-18}\left(I_x\right) \tag{5}$$

$$F_n = \{\text{ResNet-18}\left(I_1\right), \text{ResNet-18}\left(I_2\right), \ldots, \text{ResNet-18}\left(I_8\right) \tag{6}$$

$$G_x = \text{GAP}\left(F_x\right), \; G_n = \{\text{GAP}\left(F_1\right), \text{GAP}\left(F_2\right), \ldots, \text{GAP}\left(F_8\right)\}. \tag{7}$$

Next, we calculate the cosine similarity between the global feature vectors in $G_n = \{G_1, G_2, \ldots, G_8\}$ and the global feature vector G_x, as follows:

$$\text{sim}\left(G_x, G_i\right) = \frac{G_x \cdot G_i}{\|G_x\|\|G_i\|} \quad \forall i \in \{1, 2, \ldots, 8\}. \tag{8}$$

By calculating the cosine similarity of the global feature vectors, we can obtain the similarity ranking of other patches to the current patch to be segmented. We will divide these eight global vectors into four groups based on their similarity scores. Specifically, the first and eighth vectors will form one group, the second and seventh another, the third and sixth a third, and the fourth and fifth a fourth. After that, the two global vectors within each group will be added together pixel by pixel. The dimensions of obtained four global feature vectors are expanded to $512 \times (H/8) \times (W/8)$, and they are concatenated along the channel dimension. Subsequently, a 3×3 convolution is applied to reduce the channel dimension back to 512, yielding the multipatch feature G. This multipatch feature G then serves as the input to the multipatch perception branch. In the above process, the complementary aggregation of high and low similarity features enhances the robustness of feature representation, providing effective input for the multipatch perception branch.

C. Knowledge-Guided Multipatch Feature Filtering Module

The KG-MFFM is a crucial component aimed at integrating and refining multipatch features guided by domain-specific knowledge in the field of remote sensing. This module leverages the knowledge features from the knowledge-guided branch to selectively emphasize relevant information within the multipatch features, while simultaneously suppressing irrelevant information. The overall structure of KG-MFFM is shown in Fig. 5(a).

At each stage, the multipatch features entering the KG-MFFM first adjust their channel number and size to match the knowledge features of the current stage. Within the KG-MFFM, there are two stages of multipatch features filtering: self-filtering and guidance-filtering.

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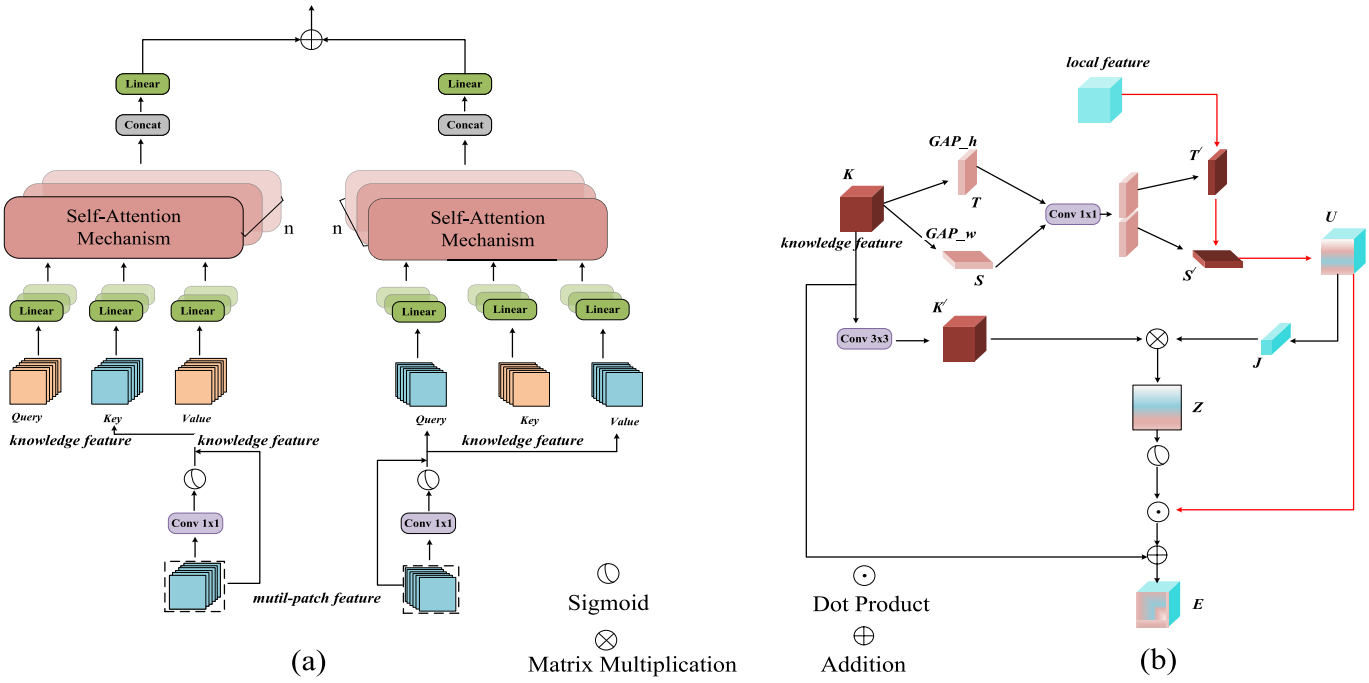


Fig. 5. Structure of the proposed two knowledge-guided modules. (a) KG-MFFM. (b) KG-LFEM.

1) Self-Filtering: We utilize the sigmoid activation function to create an attention weight map for self-filtering of multipatch features, thereby establishing a foundation for subsequent guidance-filtering. We assume the multipatch features as $G \in \mathbb{R}^{C\times H\times W}$ and the knowledge features as $K \in \mathbb{R}^{C\times H\times W}$. Initially, the multipatch features G pass through a 1×1 convolution to obtain a single-channel feature map, which is then processed by a sigmoid function to generate an attention map MAP $\in \mathbb{R}^{1\times C\times W}$. Next, this attention map MAP is element-wise multiplied with the multipatch features G to obtain G', and then G' is element-wise added back to G. The aforementioned process is expressed as follows:

$$\text{MAP} = \sigma\left(\text{Conv}_{1\times 1}(G)\right) \tag{9}$$

$$G' = G \odot \text{MAP} \tag{10}$$

where $\text{Conv}_{1\times 1}$ represents 1×1 convolution, σ represents the sigmoid function, and $\odot$ represents the dot product.

2) Guidance-Filtering: In the stage of guidance-filtering, we employed a multihead cross-attention mechanism [50]. As opposed to the self-attention mechanism, the cross-attention mechanism enables the interaction and fusion of different types of information. Our cross-attention mechanism has two parallel operations. The purpose of these two parallel cross-attentions is to more effectively screen out useful information that is beneficial to the current patch to be segmented, guided by the knowledge features. Specifically, in one set of cross-attention, the knowledge features serve as the query and value, while the multipatch features serve as the key. In the other set of cross-attention, the multipatch features serve as the query and value, and the knowledge features serve as the key. The expression of the two parallel cross-attention mechanisms mentioned above is as follows:

$$\text{CA}_1 = \text{softmax}\left(\frac{\mathbf{Q}_{\text{KF}}\mathbf{K}_{\text{MF}}^T}{\sqrt{d_k}}\right)\mathbf{V}_{\text{KF}} \tag{11}$$

$$\text{CA}_2 = \text{softmax}\left(\frac{\mathbf{Q}_{\text{MF}}\mathbf{K}_{\text{KF}}^T}{\sqrt{d_k}}\right)\mathbf{V}_{\text{MF}} \tag{12}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query, key, and value, respectively, d_k is the dimension of the key, MF represents the multipatch features, and KF represents the knowledge features.

The features obtained after implementing the two mutual cross-attention mechanisms are added together to obtain the final output $O \in \mathbb{R}^{C\times H\times W}$.

We drew inspiration from SAM's design philosophy when designing KG-MFFM [38]. To a certain extent, the multipatch feature can be regarded as the prompt in SAM. The encoding features of the prompt will interact with a mutual cross-attention mechanism with the encoding features of the patch to be segmented, hereby enhancing segmentation performance by leveraging the promotional information.

D. Knowledge-Guided Local Feature Enhancement Module

The MTP model pretrained on LSRS datasets possesses abundant domain knowledge. How to utilize this extensive domain knowledge to refine the local features generated by ConvNeXt is a crucial step in our proposed method. We propose KG-LFEM to achieve this process, and the overall structure of KG-LFEM is shown in Fig. 5(b).

Before executing operations, the knowledge features K from the knowledge-guided branch and the local features L from the local feature capture branch need to be adjusted to the same dimension, assumed to be $K \in \mathbb{R}^{C\times H\times W}$ and $L \in \mathbb{R}^{C\times H\times W}$, respectively. In KG-LFEM, there are two types of knowledge guidance processes—spatial information enhancement and channel information enhancement.

1) Spatial Information Enhancement: In the spatial information enhancement process, the knowledge features K undergo horizontal and vertical GAP to obtain two features

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with dimensions $T \in \mathbb{R}^{C\times H\times 1}$ and $S \in \mathbb{R}^{C\times 1\times W}$, respectively. Subsequently, S is reshaped to $C\times W\times 1$ and concatenated with T along the height dimension, followed by a 1×1 convolution to produce a new feature $Q \in \mathbb{R}^{C\times 2H\times 1}$. Then, Q is split into two new feature $T' \in \mathbb{R}^{C\times H\times 1}$ and $S' \in \mathbb{R}^{C\times 1\times W}$. Finally, the features of T' and S' are activated using the sigmoid activation function, and the local feature L is multiplied by T' and S' in horizontal and vertical directions to obtain the result U after the first knowledge guidance. The aforementioned process can be expressed as follows:

$$T = \mathrm{GAP}_{\mathrm{horizontal}}(K) \tag{13}$$

$$S = \mathrm{GAP}_{\mathrm{vertical}}(K) \tag{14}$$

$$G = \mathrm{Conv1}\times 1(\mathrm{concat}([T, \mathrm{reshape}(S)])) \tag{15}$$

$$[T', S'] = \mathrm{split}(Q) \tag{16}$$

$$U = L \odot \sigma(T')\,\sigma(S') \tag{17}$$

where $\mathrm{GAP}_{\mathrm{horizontal}}$ represents the horizontal GAP and $\mathrm{GAP}_{\mathrm{vertical}}$ represents the vertical GAP.

2) Channel Information Enhancement: In the channel information enhancement process, the enhanced local features U from the first knowledge guidance are first subjected to GAP, reshaped, and activated by the softmax function to obtain the vector $J \in \mathbb{R}^{1\times C}$. The knowledge features K pass through a 3×3 convolution and are reshaped to obtain the feature $K' \in \mathbb{R}^{C\times \mathrm{HW}}$. Then, J and K' undergo matrix multiplication to derive the weight relationship map $Z \in \mathbb{R}^{1\times \mathrm{HW}}$. Next, Z is activated by the sigmoid function, and the feature U is element-wise multiplied by this activated feature to obtain the output feature $E \in \mathbb{R}^{C\times H\times W}$ after the second knowledge guidance. The aforementioned process can be expressed as follows:

$$J = \mathrm{softmax}(\mathrm{reshape}(\mathrm{GAP}(U))) \tag{18}$$

$$K' = \mathrm{reshape}(\mathrm{Conv3}\times 3(K)) \tag{19}$$

$$Z = J \otimes K' \tag{20}$$

$$E = U \odot \mathrm{reshape}(\sigma(Z)). \tag{21}$$

After two rounds of knowledge-guided enhancement, E will be added pixel by pixel to the initial knowledge feature K to obtain the final output.

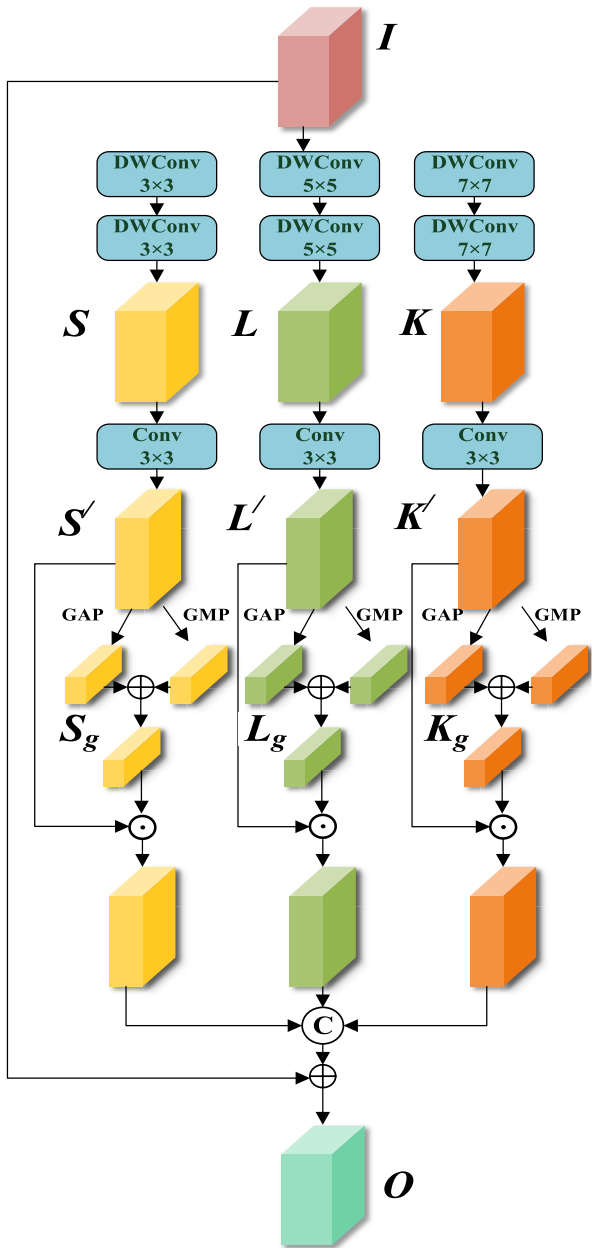


Fig. 6. Structure of GFSB.

E. Global Feature Selection Block

In our proposed method, the perception of multipatch features constructed from other patches is crucial. We have designed the GFSB as shown in Fig. 6, which serves as the foundation for constructing the multipatch perception branch to process multipatch. Specifically, the GFSB consists of two stages: the context awareness stage and the spatial feature selection stage. We assume the input feature is denoted as $I \in \mathbb{R}^{C\times H\times W}$.

1) Context Awareness Stage: In this stage, we employ two consecutive sets of parallel depthwise separable convolutions [51] with kernel sizes of 3×3, 5×5, and 7×7 to process the input feature I, yielding features $S \in \mathbb{R}^{C\times H\times W}$, $L \in \mathbb{R}^{C\times H\times W}$, and $K \in \mathbb{R}^{C\times H\times W}$. Generally, larger convolutional kernels possess a broader receptive field, enabling them to capture richer global semantic information across multiple patches. The process is formalized as follows:

$$S, L, K = \mathrm{DWConv}_{3\times 3}(I), \mathrm{DWConv}_{5\times 5}(I), \mathrm{DWConv}_{7\times 7}(I) \tag{22}$$

where $\mathrm{DWConv}_{k\times k}$ denotes depthwise separable convolution with kernel size $k\times k$.

2) Spatial Feature Selection Stage: In this stage, S, L, and K are first passed through a 1×1 convolution to reduce their channel dimensions to one-third of the original, resulting in S', L', and K', respectively. These are then subjected to GAP and GMP, with the results added pixel-wise to obtain S_g, L_g, and $K_g \in \mathbb{R}^{C\times 1\times 1}$. Next, S, L, and K are element-wise multiplied with S_g, L_g, and K_g, respectively. Immediately, S, L, and K are concatenated along the channel dimension and passed through a 3×3 convolution to produce the output

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feature $P \in \mathbb{R}^{C\times H\times W}$. Finally, the feature P is added pixel-wise to the input feature I to obtain the final output $O \in \mathbb{R}^{C\times H\times W}$.

F. Loss Function

In our research, we utilize the online hard example mining (OHEM) cross-entropy loss to enhance the learning efficiency of the network. This particular loss function focuses more intently on difficult-to-classify samples during the training phase by assigning them higher weights. The formula for OHEM cross-entropy loss is given by the following equation:

$$\mathcal{L}_{\text{OHEM}} = -\frac{1}{N}\sum_{i=1}^{N} \text{loss}_i \cdot \text{weight}_i \tag{23}$$

where loss represents the cross-entropy loss, N represents the number of ground objects, and weight represents the weight of model.

IV. Experiment

A. Dataset

1) Vaihingen Dataset: The ISPRS Vaihingen dataset[1] is a comprehensive collection featuring 33 aerial images of the urban area in Vaihingen, Germany. Each image has an average resolution of 2494 × 2064 pixels. This dataset is notably detailed, labeling six distinct categories of ground objects: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Following the official partitioning guidelines, the dataset is divided into two main parts: an 11-image training set used for model training and optimization, and a five-image test set exclusively for evaluating model performance and generalization.

2) Potsdam Dataset: The ISPRS Potsdam dataset[2] is a renowned high-resolution remote sensing image annotation dataset released in 2018. It features aerial photographs of Potsdam, Germany. This dataset classifies the same six categories of ground objects. With a spatial resolution of 0.5 m, the dataset covers an extensive area of 38.4 square kilometers. In comparison to the Vaihingen dataset, the Potsdam dataset is more comprehensive, encompassing 38 remote sensing images, each measuring 6000×6000 pixels. According to the official division, the dataset is split into 24 images for training and 14 for testing purposes.

3) UAVid Dataset: The UAVid dataset [52] is a high-resolution dataset for UAV image semantic segmentation. With resolutions of 3840 × 2160 and 4096 × 2160, it focuses on urban street scenes and contains eight semantic categories. In experiments, 150 official images are typically used. These images are padded and split into 1024 × 1024 pixel patches to facilitate model training and evaluation, serving as a crucial resource for remote sensing image segmentation research.

4) WBDS Dataset: This dataset contains two aerial images (32 507 × 15.345 pixels) from 2012 and 2016, with building labels for both years. They were cropped into 630 smaller 2,048 × 2.048 patches, split into training (378), validation (126), and test sets (126) in a 6:2:2 ratio.

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-semlabel-vaihingen/

[2]2https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-semlabel-potsdam/

B. Implementation Details

1) Experimental Settings: In our experiment, the LSRS images are cropped into patches with a resolution of 512×512 each, using a one-third overlap rate. We employed the Stochastic Gradient Descent (SGD) optimizer, configuring it with a weight decay value of 0.0005 to mitigate overfitting. The batchsize for all experiments is set to 8, and the model is trained for 200 epochs. Initiating training, we set the learning rate to 0.00006. To maintain training stability and ensure continuous, beneficial updates to the model, we implemented a learning rate decay strategy

$$\text{lr} = \text{lr}_{\text{init}} \times \left(1 - \frac{\text{current_iterations}}{\text{max_iterations}}\right)^{0.9}. \tag{24}$$

During training, MTP, ConvNextV2, and ResNet-18 utilize pretrained weight parameters provided officially. Additionally, our network adopts a two-stage training approach. The first-stage training aims to develop a model capable of efficiently segmenting independent patches, while the second stage builds upon the first to train a model that enables multipatch perception for segmentation. Specifically, the first-stage training only includes the local feature capture branch and the KG-LFEM module. The second-stage training involves the entire network structure. During testing, we employ a sliding window approach for large-scale images, and the overlap rate is 33%.

2) Evaluation Metrics: To accurately assess the efficacy of various methods, we employ three widely recognized evaluation metrics in the realm of semantic segmentation tasks. These metrics encompass the mean $F1$ score, mean intersection over union (mIoU), and overall accuracy (OA). The methodologies for computing these metrics are outlined in detail below

$$\text{OA} = \frac{N_{\text{TP}} + N_{\text{TN}}}{N_{\text{TP}} + N_{\text{FP}} + N_{\text{FN}} + N_{\text{TN}}} \tag{25}$$

$$\text{IoU} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}} + N_{\text{FN}}} \tag{26}$$

$$F_1 = \frac{2 * \left(\frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}}+\text{N}_{\text{FP}}} * \frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}}+\text{N}_{\text{FN}}}\right)}{\left(\frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}}+\text{N}_{\text{FP}}} + \frac{\text{N}_{\text{TP}}}{\text{N}_{\text{TP}}+\text{N}_{\text{FN}}}\right)} \tag{27}$$

where N_{TP} represents true positive, N_{FP} represents false positive, and N_{FN} represents false negative.

The $F1$-score is a metric that balances both precision and recall, thereby providing a comprehensive evaluation of the performance of semantic segmentation models. Meanwhile, mIoU and OA serve as objective measures to quantify the pixel-level predictive accuracy of the model.

C. Ablation Study

Our ablation experiments are organized into three parts: the ablation study on KG-LFEM, the ablation study on KG-MFFM, and the ablation study on the multipatch perception branch. For a fair comparison, we select the method that solely utilizes the local feature capture branch as our baseline.

1) Ablation Study of KG-LFEM: KG-LFEM is employed to progressively guide and enhance the representation of local features derived from the local feature capture branch, leveraging the extensive knowledge features embedded within the

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TABLE I
PROPOSED MPKG-NET EXECUTION FLOW

Require: Target patch I_x, multi-patch feature G
1: **Four-Branch Encoding:**
2: Extract features: $\begin{cases} F_{kg} = \mathcal{B}_{kg}(I_x) & \text{(Knowledge-guided)} \\ F_{local} = \mathcal{B}_{local}(I_x) & \text{(ConvNeXtV2)} \\ F_{multi} = \mathcal{B}_{multi-patch}(G) & \text{(GFSB-based)} \end{cases}$
3: **for** Each encoder stage k **do**
4: Knowledge-guided processing: $\begin{cases} F^{enh}_{local} = \text{KG-LFEM}(F_{local}, F_{kg}) \\ F^{filter}_{multi} = \text{KG-MFFM}(F_{multi}, F_{kg}) \end{cases}$
5: Feature fusion: $F^k_{output} = F^{enh}_{local} + F^{filter}_{multi}$
6: **end for**
7: **Decoding:**
8: Concatenate $\{F^k_{output}\}^4_{k=1}$
9: Upsample through decoder
10: Output segmentation mask

TABLE II
ABLATION STUDY OF KG-LFEM ON THE ISPRS VAIHINGEN DATASET

Method	params(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	0.87	88.93	76.48
Baeline+KF	411.56	3.96	89.16	78.90
Baseline+KG-LFEM(LF+KF)	420.78	4.17	**89.54**	**79.76**

TABLE III
ABLATION STUDY OF KG-LFEM ON THE ISPRS POTSDAM DATASET

Method	params(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	5.98	90.11	85.73
Baeline+KF	411.56	26.71	90.51	86.12
Baseline+KG-LFEM(LF+KF)	420.78	28.13	**90.86**	**86.88**

knowledge-guided branch in the context of remote sensing. To validate the effectiveness of KG-LFEM, we conduct comparative experiments on the two datasets, with the experimental results presented in Tables II and III.

We conduct three sets of comparative experiments: 1) baseline; 2) baseline + KF: The features of the knowledge-guided branch and the features of the local feature capture branch are added stage by stage. 3) baseline + KG − LFEM(LF + KF): The features of the local feature capture branch and the knowledge-guided branch are fused through KG-LFEM. As shown in Tables II and III, it is evident that the best performance is achieved when KG-LFEM is used for progressive knowledge-guided local feature enhancement. Specifically, on the Vaihingen dataset, the baseline exhibits an improvement of 0.23% in OA and 2.42% in mIOU after incorporating the knowledge-guided branch. Additionally, the introduction of KG-LFEM further improves OA by 0.38% and mIOU by 0.86% without causing a significant increase in computational complexity. Similar significant improvements are also observed on the Potsdam dataset.

2) Ablation Study of KG-MFFM: To effectively screen and filter the global features provided from other patch images, KG-MFFM is designed to utilize the features from the knowledge-guided branch to guide the features from the multipatch feature perception branch, achieving stage-wise screening and filtering of multipatch features. To validate the effectiveness of KG-MFFM, we conduct ablation experiments

TABLE IV
ABLATION STUDY OF KG-MFFM ON THE ISPRS VAIHINGEN DATASET

Method	params(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	0.87	88.93	76.48
Baseline+MF	118.22	1.02	87.01	73.09
Baseline+(MF+KF)	436.47	4.52	88.23	73.72
Baseline+KG-MFFM(MF+KF)	438.57	4.71	**91.11**	**83.25**

TABLE V
ABLATION STUDY OF KG-MFFM ON THE ISPRS POTSDAM DATASET

Method	params(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	5.98	90.11	85.73
Baseline+MF	118.22	6.86	88.71	82.30
Baseline+(MF+KF)	436.47	30.54	89.10	83.99
Baseline+KG-MFFM(MF+KF)	438.57	31.80	**91.37**	**89.11**

TABLE VI
ABLATION STUDY ON THE INTERNAL COMPONENTS OF THE KG-LFEM AND KG-MFFM DATASETS

Method	OA(%)	mIOU(%)
KG-LFEM$_{sim}$	89.24	79.27
KG-LFEM$_{cim}$	89.29	79.19
KG-LFEM	**89.54**	**79.76**
KG-MFFM$_{sf}$	88.78	76.73
KG-MFFM$_{gf1}$	91.01	82.86
KG-MFFM$_{gf2}$	90.83	82.61
KG-MFFM	**91.11**	**83.25**

on the ISPRS Vaihingen and Potsdam datasets, with the experimental results presented in Tables IV and V. There are three sets of comparative experiments in Tables IV and V: 1) baseline; 2) baseline + MF: the features of the multipatch perception branch and the features of the local feature capture branch are added stage by stage: 3) baseline + (MF + KF): the features of the multipatch perception branch and the knowledge-guided branch are added stage by stage, and then the features of the local feature capturebranch are added stage by stage; 4) Baseline+KG−MFFM(MF+KF): the features of the multipatch perception branch and the knowledge-guided branch are fused through KG-MFFM, and then the features of the local feature capture branch are added stage by stage. It can be observed that when only the multipatch perception branch is added, OA decreases compared to the baseline. This is because the unfiltered multipatch features contain a significant amount of redundant information, which can interfere with the semantic segmentation of the current patch image to some extent. When we use KG-MFFM to screen and filter multipatch information, the model performs best without adding extra computational burden (the number of parameters only increases by about 2M).

Additionally, we conducted ablation study on the internal components of KG-LFEM and KG-MFFM on the Vaihingen dataset, as shown in Table VI. KG-LFEM$_{\text{sim}}$

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TABLE VII
ABLATION STUDY OF THE MULTIPATCH PERCEPTION BRANCH ON THE ISPRS VAIHINGEN DATASET

Method	param(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	0.87	88.93	76.48
without MPPB	411.56	3.86	89.16	78.90
with MPPB	484.16	5.15	**91.68**	**83.45**

TABLE VIII
ABLATION STUDY OF THE MULTIPATCH PERCEPTION BRANCH ON THE ISPRS POTSDAM DATASET

Method	param(m)	time(s)	OA(%)	mIOU(%)
Baseline	90.95	5.98	90.11	85.73
without MPPB	411.56	26.06	90.51	86.12
with MPPB	484.16	35.76	**91.81**	**89.35**

and KG-LFEM$_{cim}$ represent KG-LFEM that retains only spatial information enhancement and channel information enhancement, KG-MFFM$_{sf}$KG-LFEM$_{gf1}$ and KG-LFEM$_{gf2}$ represent KG-MFFM that retains only self-filtering, left side cross-attention mechanism, and right side cross-attention mechanism.

3) Ablation Study of the Multipatch Perception Branch: Compared to the majority of LSRS image semantic segmentation methods, a significant highlight of our proposed method is capturing features from relevant patch images through multipatch perception branch. To validate the effectiveness of the multipatch perception branch in our method, we conduct the following experiments for verification on the ISPRS Vaihingen and Potsdam datasets: 1) baseline; 2) without MPPB: retain the local feature capture branch and the knowledge-guided branch; and 3) with MPPB: Complete model.

From Tables VII and VIII, we observe that when the multipatch perception branch is employed, our method achieves improvements of 2.75% and 1.70%, respectively, in OA compared to the baseline on the two datasets. This adequately demonstrates that perceiving multiple patches can provide more global semantic information for the currently targeted patch to be segmented.

D. Comparison With State-of-the-Art Methods

To validate the effectiveness of our proposed method, we have compared it with other mainstream approaches. The selected comparison methods are categorized into two major groups: CNN-based and Transformer-based.

The CNN-based methods include simple encoder–decoder architectures such as FCN, UNet, and SegNet; context information enhancement methods like DeepLabV3+, PSPNet, and HRNet; attention mechanism-based methods such as DANet, CCNet, and OCRNet. Additionally, we have selected some typical methods: BiSeNetV2, HCANet, and PIDNet. The Transformer-based methods encompass Segformer, Topformer, TransUNet, and EMRT. Additionally, we also select some methods based on knowledge-guided or multipatch perception for comparison: SAMUNet, MTP, and PTRSegNet.

Tables X–XII present the specific experimental results. Experimental results on the ISPRS Vaihingen dataset reveal that early encoder–decoder architecture methods, such as FCN, UNet, and SegNet, exhibit suboptimal segmentation performance. Subsequently, methods focusing on enlarged receptive fields achieved significant improvements, with DeepLabV3+ notably outperforming previous approaches in terms of OA, mIOU, and other metrics. This advancement is primarily attributed to the ability to capture richer global information from ground objects through an expanded receptive field. Similarly, attention mechanism-based methods, including DANet, CCNet, and OCRNet, establish global relationships by constructing various spatial and channel attention mechanisms, thereby substantially enhancing segmentation performance. In experiments involving Transformer-based methods, we observe that their overall performance is inferior to CNN-based methods. This may primarily be due to the need for further design and innovation of the original Transformer architecture to address complex remote sensing scenarios. We also found that knowledge-guided methods such as SAMUNet and MTP have achieved significant improvements in segmentation performance compared with previous methods. This is mainly attributed to the large-scale pretraining, which endows the model with stronger domain task performance capabilities. Finally, our proposed MPKG-Net has achieved significant improvements compared with both CNN-based and Transformer-based methods. For example, on the ISPRS Potsdam dataset, MPKG-Net achieves the highest $F1$-scores for four ground objects, with an OA 0.14% higher and an mIOU 0.41% higher than the second-ranked AerialFormer. These improvements are mainly attributed to MPKG-Net's ability to acquire richer global information through multipatch perception and leverage semantic information more effectively via the knowledge-guided architecture to enhance segmentation performance. Similar conclusions are drawn on the ISPRS Vaihingen and UAVid datasets. In comparison with some current state-of-the-art methods, our method also exhibits distinct advantages. AerialFormer, leveraging the strengths of combining a Transformer and CNN to retain low-level high-resolution features, has achieved outstanding results on the ISPRS Potsdam dataset. In comparison with it, our method has taken the lead in terms of mIOU and OA. LSKNet is a lightweight foundation model. Although it has achieved remarkable performance on the Vaihingen dataset, our method has obtained more competitive results on the larger-scale datasets Potsdam and UAVid, demonstrating the robustness and generalization ability of our method. PTRSegNet bears certain similarities to our method, as it also captures global semantic information by processing multiple patches. However, on this basis, our method, with the guidance of knowledge, has achieved obvious advantages on all three datasets. In comparison with some current state-of-the-art methods, our method also exhibits distinct advantages. AerialFormer, leveraging the strengths of combining a Transformer and CNN to retain low-level high-resolution features, has achieved outstanding results on the ISPRS Potsdam dataset. In comparison with it, our method has taken the lead in terms of mIOU and OA. LSKNet is a lightweight foundation model. Although it has achieved

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TABLE IX
EXPERIMENTAL RESULTS ON THE ISPRS POTSDAM DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	mean F_1(%)	mIoU(%)	OA(%)
FCN [11]	91.85	95.94	85.82	87.21	88.86	89.94	84.38	89.69
UNet [13]	91.14	94.38	85.01	87.33	87.99	89.17	83.88	87.71
SegNet [12]	91.62	94.71	84.69	86.98	94.46	90.49	82.88	88.60
DeeplabV3+ [15]	92.95	95.88	87.62	88.15	96.02	92.12	84.32	90.88
PSPNet [20]	91.78	96.08	85.81	87.39	88.14	89.84	84.12	89.61
HRNet [18]	93.44	96.86	87.02	88.11	96.24	92.33	86.17	90.98
DANet [21]	92.59	96.49	86.29	87.62	89.30	90.46	84.92	90.20
CCNet [17]	91.92	95.97	86.33	87.31	89.17	90.14	84.66	90.07
OCRNet [32]	93.01	95.97	86.39	87.88	95.81	91.81	86.00	90.55
BiSeNet [53]	92.65	96.18	86.52	87.42	95.06	91.57	87.19	90.37
PIDNet [54]	91.77	95.98	86.17	86.96	91.88	90.55	85.02	89.43
Segformer [35]	90.65	94.21	83.28	82.94	90.65	88.35	82.34	87.34
Topformer [36]	90.03	93.78	82.95	82.45	90.06	87.85	82.01	87.12
TransUNet [55]	90.17	93.35	83.31	83.68	91.61	88.42	79.52	87.10
EMRT [56]	90.87	94.86	84.12	85.23	90.32	89.08	80.54	88.12
AerialFormer [57]	95.4	**98.0**	89.6	89.7	97.4	**94.00**	89.0	93.8
MTP [26]	91.11	95.02	85.46	86.24	92.18	90.00	83.45	89.17
SAMUNet [58]	92.14	96.07	85.97	86.71	92.11	90.60	84.04	90.56
PTRSegNet [25]	93.49	96.99	87.01	87.07	94.18	91.75	86.21	92.56
MPKG-Net+bce	94.84	96.71	89.03	88.99	96.51	93.22	88.89	93.49
MPKG-Net(Ours)	**95.42**	97.58	**89.70**	**89.80**	**97.47**	93.98	**89.41**	**93.94**

TABLE X
EXPERIMENTAL RESULTS ON THE ISPRS VAIHINGEN DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	mean F_1(%)	mIoU(%)	OA(%)
FCN [11]	90.44	94.56	80.99	87.52	61.87	83.08	73.55	88.41
UNet [13]	87.88	91.90	81.44	87.96	75.55	84.94	75.96	88.16
SegNet [12]	88.63	92.04	78.90	88.01	74.34	84.38	74.85	88.67
DeepLabV3+ [15]	90.55	95.01	82.31	87.98	77.51	86.67	80.99	89.75
PSPNet [20]	90.87	95.67	81.57	88.32	72.67	85.82	80.84	89.72
HRNet [18]	92.13	95.98	83.74	87.99	88.04	89.58	81.88	90.64
DANet [21]	90.09	95.01	81.33	88.56	74.29	85.86	80.96	89.66
CCNet [17]	91.05	95.42	80.95	88.06	75.62	86.22	80.55	89.83
OCRNet [32]	90.99	95.92	81.20	88.71	80.01	87.37	81.13	90.16
HCANet [59]	92.47	95.01	84.21	89.37	84.02	89.02	81.03	90.33
DLNet [60]	91.29	95.25	78.78	87.91	79.68	86.58	76.99	88.71
BiSeNetV2 [53]	91.17	96.02	81.12	88.01	77.03	86.67	80.96	89.94
PIDNet [54]	90.34	94.87	80.21	87.65	76.56	85.92	79.88	89.02
MTP [26]	92.77	95.00	84.50	88.99	85.12	89.28	81.58	90.30
SAMUNet [58]	92.99	95.13	84.56	89.38	85.01	89.41	81.75	90.34
Segformer [35]	87.89	90.88	81.02	88.01	57.21	81.00	71.35	87.53
Topformer [36]	87.45	90.45	80.48	87.63	54.65	80.13	71.02	86.31
TransUNet [55]	86.73	88.91	81.07	88.23	55.66	80.12	70.95	86.17
EMRT [56]	89.16	92.68	78.69	86.54	80.12	85.44	74.96	86.97
PTRSegNet [25]	91.89	94.87	85.01	89.29	85.07	89.23	81.46	90.29
MPKG-Net+bce	93.11	96.99	85.42	89.76	89.45	90.95	83.78	92.04
MPKG-Net(Ours)	**93.69**	**97.11**	**85.91**	**90.02**	**89.77**	**91.30**	**84.01**	**92.34**

remarkable performance on the Vaihingen dataset, our method has obtained more competitive results on the larger-scale datasets Potsdam and UAVid, demonstrating the robustness and generalization ability of our method. PTRSegNet bears certain similarities to our method, as it also captures global semantic information by processing multiple patches. However, with the guidance of knowledge, our method has achieved obvious advantages on all datasets. In addition, we compared the effectiveness of using cross-entropy loss and OHEM loss. On the above three datasets, OHEM loss can focus on learning difficult categories, thus improving the F1 score of each ground object. Our method also outperforms other advanced methods on the WBDS dataset.

Figs. 7 and 8 show the visualization of the results. In these figures, we indicate the areas worth paying attention to with red circles. In addition, to demonstrate the effectiveness

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TABLE XI
EXPERIMENTAL RESULTS ON THE UAVID DATASET

Method	Clutter	Building	Road	Tree	Low Veg.	Mov. Car	Sta. Car	Human	mIoU(%)	OA(%)
FCN [11]	63.91	84.72	76.51	78.32	61.88	65.87	45.54	22.26	62.38	84.55
UNet [13]	61.80	82.94	75.15	77.27	62.03	59.59	29.98	18.62	58.42	83.43
DeepLabV3+ [15]	68.86	87.62	82.22	79.76	65.88	69.86	55.39	26.07	66.96	86.94
PSPNet [20]	68.89	87.96	82.20	81.07	65.69	70.30	56.31	24.67	67.13	87.28
DANet [21]	64.85	85.88	77.94	78.29	61.47	59.64	47.44	9.14	60.58	84.99
BiSeNetV2 [53]	61.18	81.62	77.11	75.97	61.30	66.36	38.51	15.40	59.68	83.10
SegNet [12]	65.62	85.89	79.23	78.78	63.73	68.94	52.10	19.29	64.20	85.54
MFNet [61]	69.66	88.63	82.51	81.31	66.42	73.21	60.59	27.44	68.72	87.63
Segformer-B5 [35]	62.10	82.76	75.10	77.13	61.98	60.10	29.55	18.43	58.39	83.16
MTP [26]	66.32	86.74	78.95	79.12	64.83	68.02	54.67	24.91	65.45	88.28
SAMUNet [58]	66.95	87.02	79.33	79.45	66.14	68.71	57.12	26.18	66.36	88.33
LSKNet [62]	69.60	84.80	82.90	80.90	65.50	**76.80**	64.90	31.80	70.00	-
PTRSegNet [25]	68.74	83.97	82.56	81.11	65.75	75.61	**64.95**	**30.81**	69.19	89.07
MPKG-Net+bce	70.24	89.05	82.97	81.89	66.91	74.98	63.21	28.76	69.75	89.78
MPKG-Net(Ours)	**70.53**	**89.31**	**83.55**	**82.54**	**67.23**	75.85	63.58	28.89	**70.19**	**90.10**

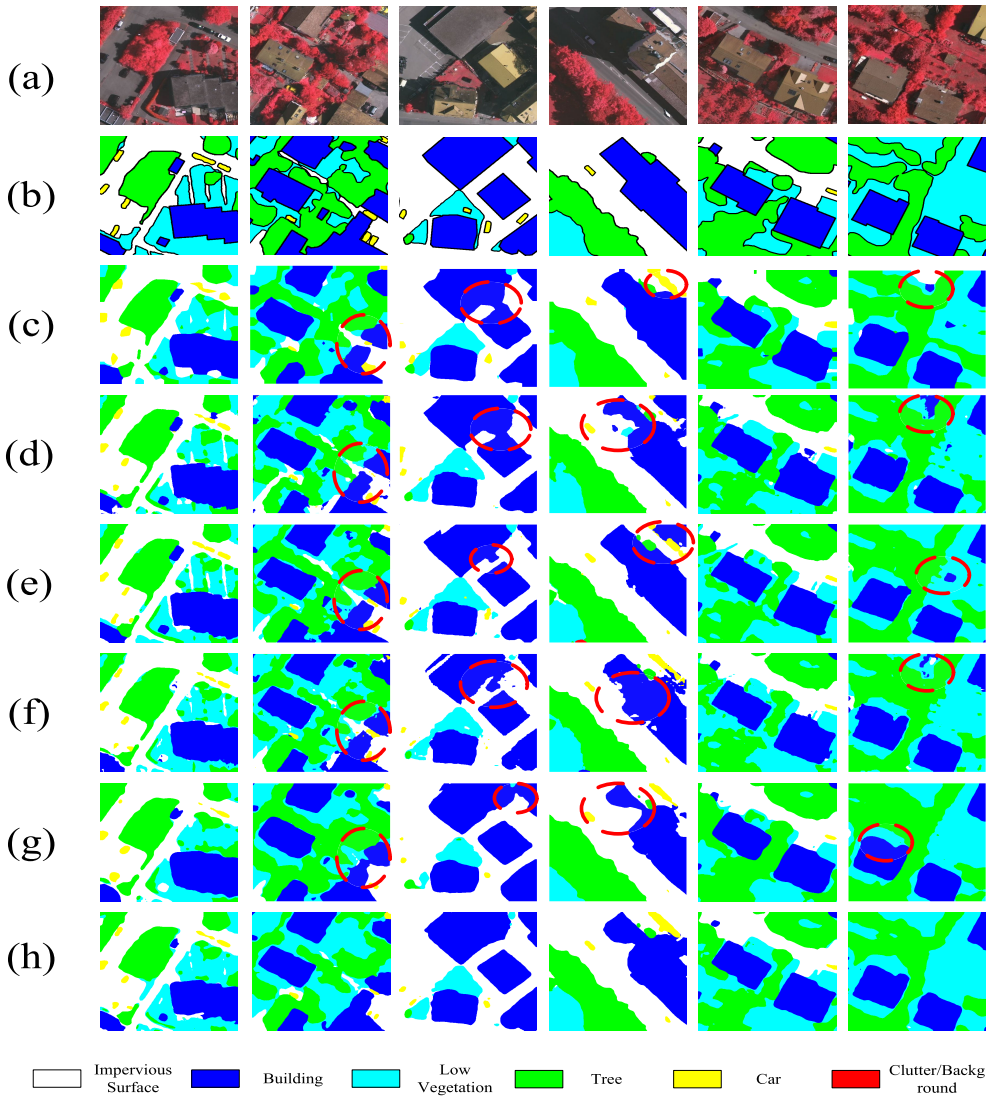


Fig. 7. Comparison of segmentation results on the ISPRS Vaihingen dataset. (a) Input image. (b) Ground truth. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) Segformer. (g) HRNet. (h) MPKG-Net (ours).

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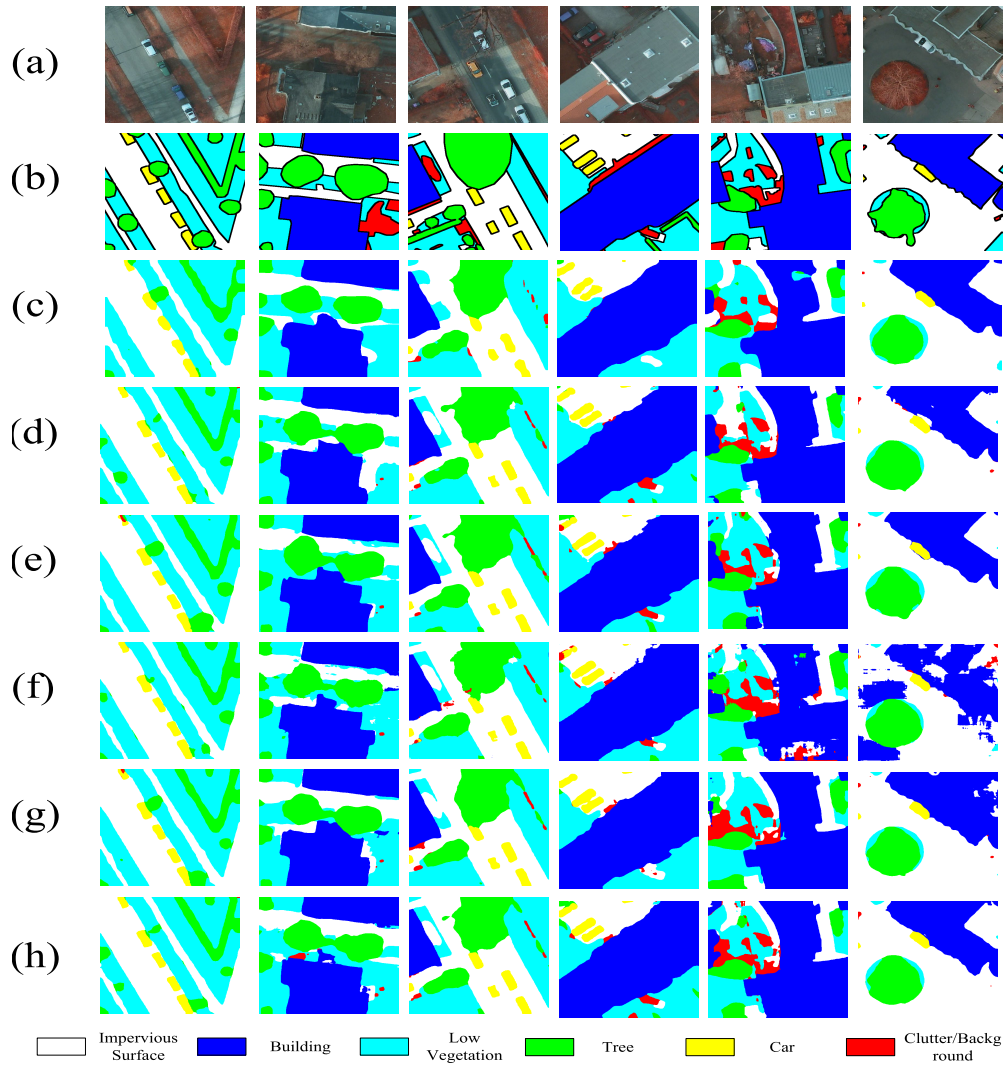


Fig. 8. Comparison of segmentation results on the ISPRS Potsdam dataset. (a) Input image. (b) Ground truth. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) Segformer. (g) HRNet. (h) MPKG-Net (ours).

TABLE XII
EXPERIMENTAL RESULTS ON THE WBDS DATASET

Method	mF1(%)	mIOU(%)
UNet	92.97	87.34
DANet	89.05	81.34
SegNet	93.72	90.99
DeepLabV3+	95.56	91.71
HRNet	95.78	92.73
MTP	91.78	88.86
MACUnet	85.53	82.20
CMTFNet	86.21	92.59
PTRSegNet	95.54	91.46
MPKG-Net	**97.36**	**94.94**

TABLE XIII
COMPARISON OF VARIOUS LSRS SEMANTIC SEGMENTATION METHODS ON THE ISPRS VAIHINGEN DATASET

Type	Method	OA (%)
CNN-based	DSPCANet [63] (JSTARS21)	87.32
	HCANet [59] (GRSL21)	90.33
	SBANet [40] (TGRS21)	90.59
	SSAtNet [39] (TGRS21)	88.01
	CEN [64] (GRSL22)	90.90
	MFANet [65] (GRSL22)	90.60
Transformer-based	DPFANet [66] (TGRS23)	91.06
	EMRT [56] (TGRS23)	88.12
	GASOT-Net [67] (TGRS24)	91.35
Proposed Method	MPKG-Net	**92.34**

of high-resolution image segmentation more intuitively, we also provide complete resolution segmentation results on two datasets, as shown in Figs. 9 and 10. From the comparison of the segmentation results, we can see that our proposed MPKG-Net achieves more complete segmentation and accurate boundary contours in the segmentation of ground objects compared to other methods. This clearly demonstrates the superiority of our proposed MPKG-Net. In addition, our

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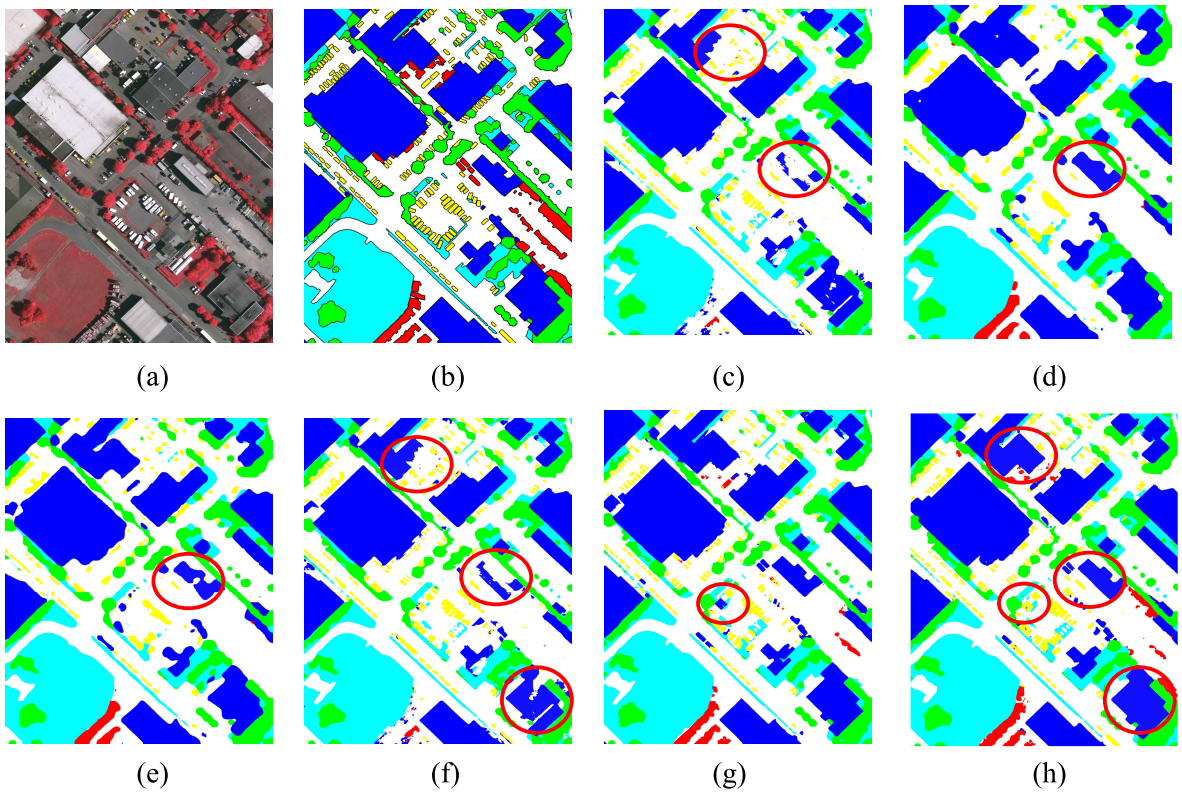


Fig. 9. Comparison of segmentation of LSRS results on the ISPRS Vaihingen dataset. (a) Input. (b) Label. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) TransUNet. (g) HRNet. (h) MPKG-Net (ours).

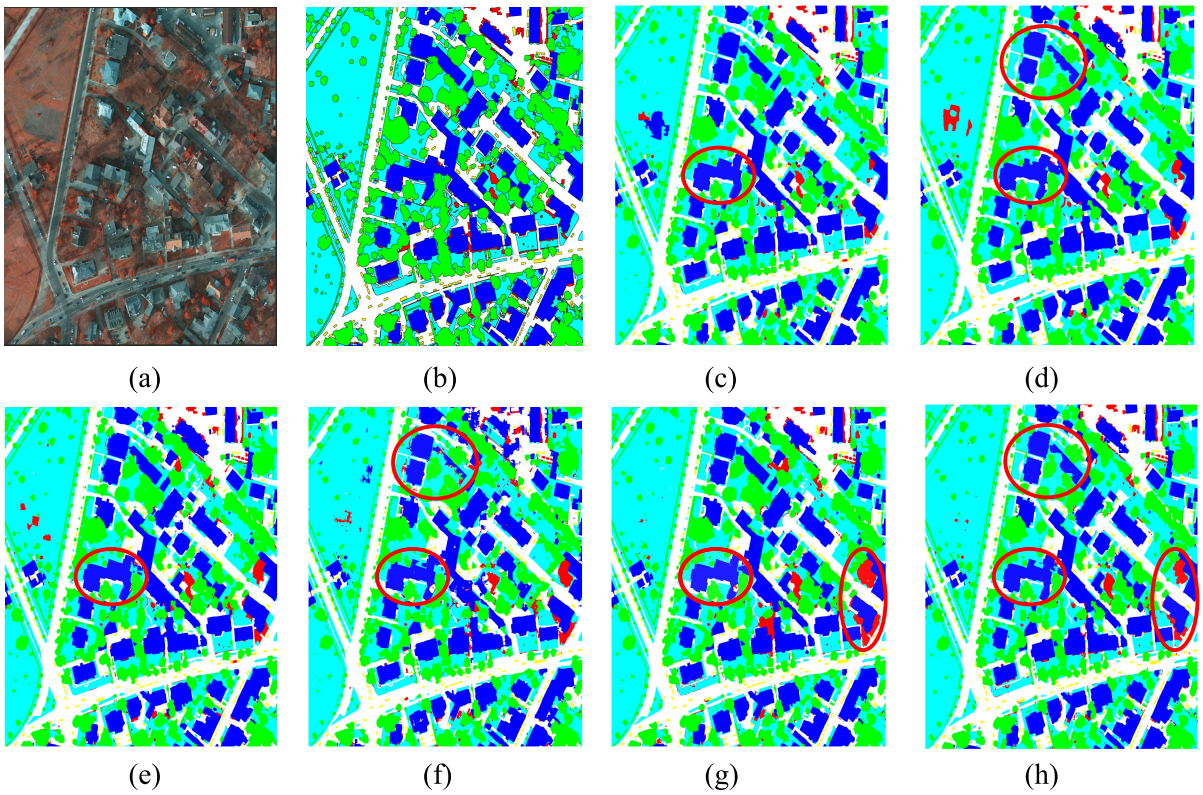


Fig. 10. Comparison of segmentation of LSRS results on the ISPRS Potsdam dataset. (a) Input. (b) Label. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) TransUNet. (g) HRNet. (h) MPKG-Net (ours).

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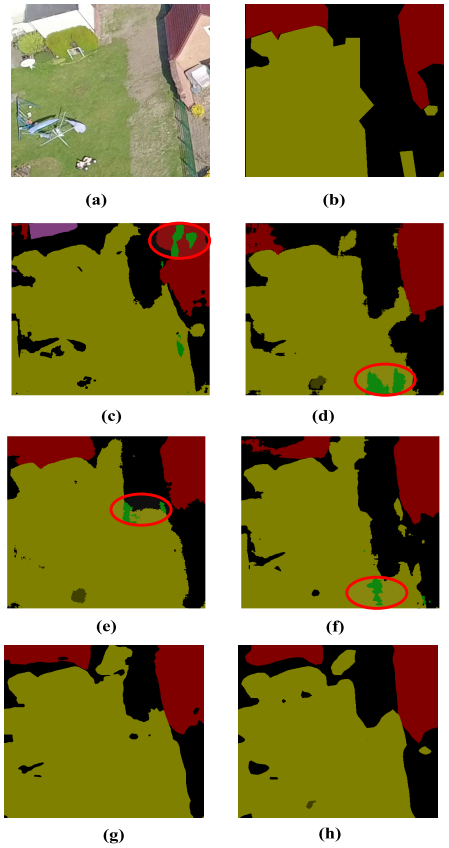


Fig. 11. Comparison of segmentation of results on the UAVid dataset. (a) Input. (b) Label. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) MTP. (g) HRNet. (h) MPKG-Net.

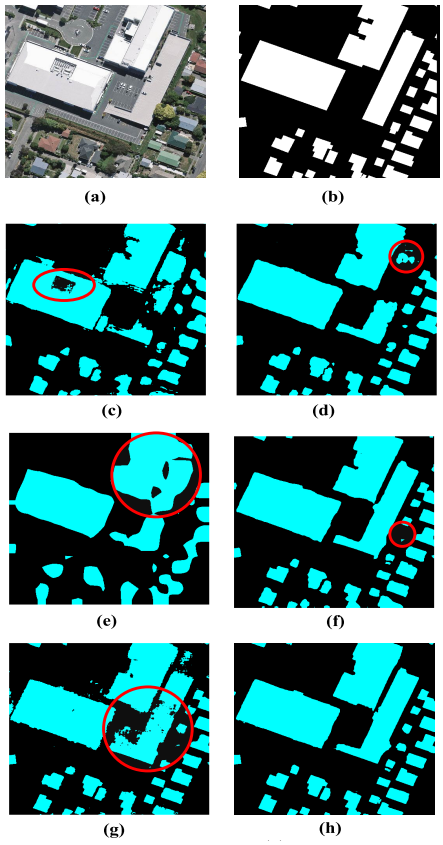


Fig. 12. Comparison of segmentation of results on the WBDS dataset. (a) Input. (b) Label. (c) UNet. (d) DeepLabV3+. (e) DANet. (f) MTP. (g) HRNet. (h) MPKG-Net.

TABLE XIV

COMPARISON OF VARIOUS LSRS SEMANTIC SEGMENTATION METHODS ON THE ISPRS POTSDAM DATASET

Type	Method	OA (%)
CNN-based	DSPCANet [63] (JSTARS21)	90.13
	HCANet [59] (GRSL21)	90.60
	MFANet [65] (GRSL22)	90.88
Transformer-based	DPFANet [66] (TGRS23)	91.73
	EMRT [56] (TGRS23)	86.97
	FTransUNet [43] (TGRS24)	90.01
	TCNet [42] (TGRS24)	90.28
	GASOT-Net [67] (TGRS24)	91.44
	AerialFormer [57] (Remote Sensing24)	93.8
	LSKNet [62](IJCV24)	92.00
	PTRSegNett [25](JSTARS24)	92.56
Proposed Method	MPKG-Net	**93.94**

TABLE XV

COMPUTATIONAL COMPLEXITY OF EACH BRANCH OF MPKG-NET

Method	Params (M)	FLOPs (GMac)
The local feature capture branch	88.54	80.34
The knowledge-guided branch	317.33	316.52
The multi-patch perception branch	29.62	109.97
The multi feature fusion branch	42.48	121.73

proposed MPKG-Net still achieved significant advantages compared to the representative LSRS semantic segmentation methods in recent years, as shown in Tables XIII and XIV.

By comparing the visualization results in Figs. 7–12, we can draw the following conclusions.

1) For the segmentation of similar ground objects, MPKG-Net demonstrates superior performance. On the ISPRS Vaihingen and Potsdam datasets, impervious surface and buildings, as well as clutter, low vegetation, and trees, often share similar appearance characteristics. Previous methods as DeepLabV3+, often struggle to effectively distinguish these categories.
2) Improved perception capability for small targets. LSRS images contain many small-scale objects, such as cars. Compared to other representative methods, our approach significantly enhances the segmentation performance for small targets (e.g., cars) through multipatch global joint perception.
3) Enhanced clarity in the segmentation of ground object boundaries. In addition to small-scale objects, LSRS images also include many large-scale objects with continuity, such as Building. Most previous segmentation methods process independent patches separately, which can easily lead to discontinuous and unclear boundaries. Thanks to multipatch perception, our method achieves noticeably sharper boundaries when segmenting these large-scale ground objects.

E. Discussion

1) Comparison of Computational Complexity: We conduct computational complexity analysis and comparisons on

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TABLE XVI
COMPARISON OF COMPUTATIONAL COMPLEXITY OF VARIOUS METHODS

Method	Traing Params (M)	FLOPs (GMac)	FPS
DeepLabV3+	45.67	56.37	19.54
PSPNet	65.58	255.62	13.45
LSKNet	17.22	89.05	21.56
PTRSegNet	14.77	64.81	17.09
AerialFormer	117.03	181.67	8.91
SAMUNet	377.64	525.34	7.74
MTP	3.35	56.78	22.11
MPKG-Net	171.16	321.04	9.05

TABLE XVII
COMPARISON OF TEST RESULTS OF DIFFERENT METHODS UNDER DIFFERENT OVERLAP RATES ON THE VAIHINGEN DATASET

overlap rate	0%		33%		50%		75%	
	time(s)	OA	time(s)	OA	time(s)	OA	time(s)	OA
HRNet	1.87	90.60	3.59	90.64	5.98	90.67	18.82	90.55
SAMUNet	3.30	90.27	6.33	90.34	10.54	90.42	33.21	90.65
LKSNet	1.15	89.88	2.21	90.07	3.68	90.15	11.59	90.17
AerialFormer	3.05	90.64	5.98	90.77	10.04	90.66	30.71	90.75
MPKG-Net	3.01	92.28	5.76	92.34	9.59	92.31	30.21	92.22

TABLE XVIII
COMPARISON OF TEST RESULTS OF DIFFERENT METHODS UNDER DIFFERENT OVERLAP RATES ON THE POTSDAM DATASET

overlap rate	0%		33%		50%		75%	
	time(s)	OA	time(s)	OA	time(s)	OA	time(s)	OA
HRNet	10.76	90.56	24.20	90.98	39.52	91.09	144.62	91.20
SAMUNet	18.98	90.00	42.70	90.56	69.72	90.48	255.16	90.74
LKSNet	6.62	90.04	14.90	90.34	24.33	90.49	89.06	90.55
AerialFormer	17.81	92.94	39.55	93.51	64.67	93.47	235.81	93.70
MPKG-Net	17.27	93.41	38.85	93.94	63.43	93.90	232.13	93.82

selected representative algorithms, as shown in Table XVI. The metrics we selected for comparison include the number of training parameters, floating-point operations per second (FLOPs), and frames per second (FPS). Earlier representative algorithms, such as DeepLabV3+ and PSPNet, primarily adopt ResNet-based encoder architectures, resulting in relatively low computational complexity but suboptimal performance on LSRS image segmentation. In recent years, with the rapid development of foundational vision models, an increasing number of algorithms have attempted to employ models with larger parameter sizes to enhance performance in visual tasks. For instance, SAMUnet leverages SAM's robust segmentation capabilities to process remote sensing images, though this comes at the cost of significantly increased computational complexity. While the lightweight foundation model LSKNet boasts lower computational demands, its segmentation performance across multiple datasets falls short compared to our proposed method. In our method, the knowledge-guided branch remains frozen during training, thus not imposing additional training burdens. Moreover, the proposed knowledge-guided modules and the GFSB are of low parameter count, which will not significantly increase the inference time. In addition, we also compared the computational complexity of each branch in MPKG-Net, as shown in Table XV.

2) Comparison of Test Results of Different Methods Under Different Overlap Rates: We evaluate the segmentation results of LSRS images using sliding windows with different overlap ratios on the Vaihingen and Potsdam datasets. Tables XVII and XVIII present the inference time and corresponding performance under various overlap ratios. The average resolution of the test images in Vaihingen is 2560 $\times$ 2083, while that of the Potsdam test images is 6000 $\times$ 6000. The following conclusion can be drawn from the results.

1) Initially, as the overlap ratio increases, the performance of all methods improves. This is primarily because a higher overlap ratio provides additional contextual information, particularly along object boundaries.
2) With further increases in the overlap ratio, the performance of some methods stabilizes or slightly declines, likely because the models have reached their upper performance limits.
3) With its knowledge-guided multipatch perception, MPKG-Net achieves highly competitive performance even with relatively small overlap ratios, which require less inference time.

V. CONCLUSION

In this article, we propose an MPKG-Net for the semantic segmentation of LSRS images. MPKG-Net, leveraging the multipatch perception branch built upon the GFSB, is capable of progressively perceiving features of patches related to the target patch to be segmented. Furthermore, with the assistance of the knowledge-guided branch, along with the KG-LFEM and the KG-MFFM, it achieves enhancement of local features from the local feature capture branch and filtering of multipatch features. In the multifeature fusion branch, features from various encoding stages are integrated and concatenated during the decoding stage to produce the final prediction. Extensive experiments have been conducted on two datasets—ISPRS Vaihingen and Potsdam. The results demonstrate that our proposed MPKG-Net outperforms the majority of current methods for LSRS images' semantic segmentation.

In our future work, we will pay more attention to the global semantic connections among patches within LSRS images. Additionally, we aim to apply the architecture proposed in this article across various remote sensing tasks by integrating multimodal information, thereby contributing to the advancement of remote sensing interpretation.

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